

Yina Kim

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Research Interest: physical human-robot interaction, sensor-based robot perception

EDUCATION

Sookmyung Women's University, Seoul, Korea

Mar. 2023 – Feb. 2027

- B.S. in Mechanical System Engineering, Intensive Major
- Overall GPA: 3.92/4.5

Rhine-Waal University of Applied Sciences, Germany

Sep. 2025 – Feb. 2026

- Exchange Student in Mechanical Engineering and Mechatronics

RESEARCH EXPERIENCE

Undergraduate Research Intern, SPRINT Lab, Sookmyung Women's University

Mar. 2025 – Aug. 2025;

Mar. 2026 – Present

- Conducted ROS/Docker-based robot perception and SLAM experiments at SPRINT Lab, including LiDAR mapping, sensor-based evaluation, and FAST-LIVO2-based pipe odometry research.

PROJECTS

2D LiDAR SLAM Localization

2025

Sookmyung Women's University

- Implemented LiDAR-based 2D SLAM and path planning on a TurtleBot3 Burger using ROS 2, LiDAR MID360, and Cartographer, and improved mapping stability by resolving tracking-frame-related map rotation.

Robot Arm Handover Demo

2025

Sookmyung Women's University

- Developed ROS 2 Python control code for an OpenManipulator-X handover demonstration, focusing on grasping, transfer, motion repeatability, and human-facing interaction timing.

FAST-LIVO2-based Pipe Odometry

2026 – Present

Industry–University Collaborative Project, Sookmyung Women's University

- Worked on a FAST-LIVO2-based pipe inspection SLAM pipeline for localization in cylindrical environments, with a plan to draft a paper by the end of August 2026.

LEADERSHIP

Robotics Academic Society <RIME>, Sookmyung Women's University

Mar. 2024 – Present;

Vice President from Mar. 2026

- Led robotics study sessions and MATLAB mentoring, supporting project planning and technical troubleshooting.

HONORS AND AWARDS

- Academic Excellence Scholarship, Jan. 2024
- Special University Scholarship for Academic Support, Feb. 2024, Feb. 2026
- Career Development Scholarship, Jul. 2024, Feb. 2026

QUALIFICATIONS AND SKILLS

- Qualifications: TOEFL iBT 88
- Skills: ROS 1/2, Docker, Ubuntu/Linux, Python, C++, MATLAB/Simulink, OpenCV, Scikit-learn